



S.P.M College, Udantpuri

Bachelor Of Computer Application (BCA)

Part -2 (Paper-4)

– Hira Kumar

Computer Networking



Data Communication

❖ Communication Media / Transmission Media / Communication Line

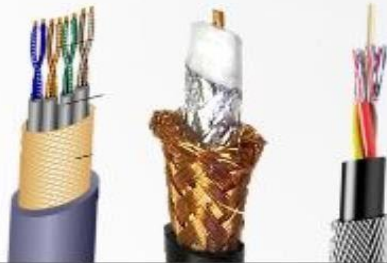
- The transmission medium is the means by which data/signal is transmitted/carried from one location to another location.
- A part of physical layer in networking / Internet.
- A transmission medium can be defined as anything that can carry analog or digital information from a source to a destination.
- Any message can be sent in digital form as data that can be translated into binary digits(0,1).The signal is encoded/encrypted with these digits. So,data is sent to a network, the network divides into packets, and the packets are formed by the network.
- Just like A mail carrier truck or train/aeroplane could be the delivery medium for a written letter/goods. Similarly,data transfer one nodes to another nodes.

Communication Channels

(Communication Media or Transmission Media)

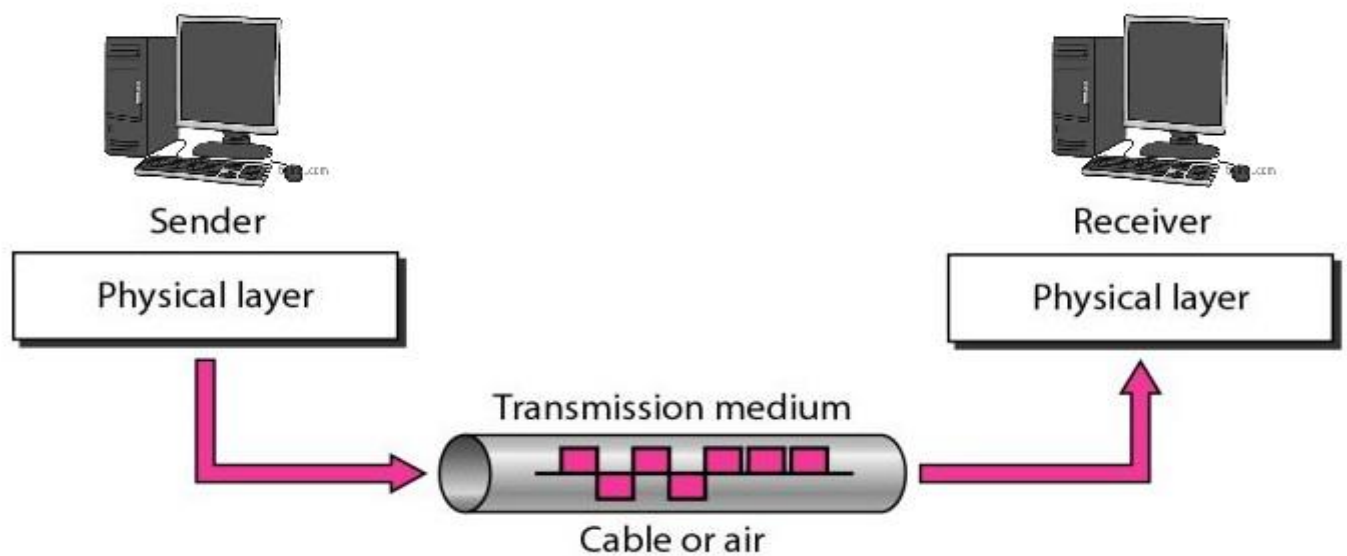
Guided Media
(wired or bounded media)

Twisted Pair Coaxial Fibre Optics

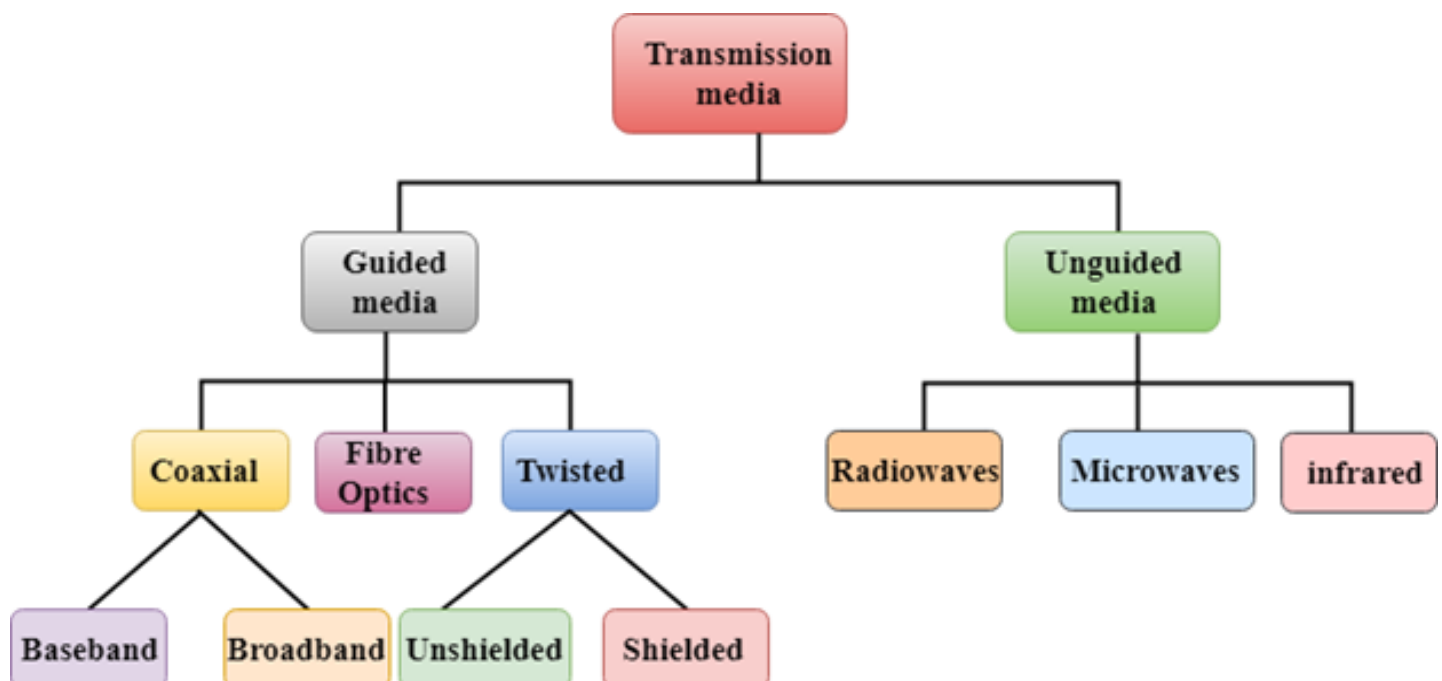


Unguided Media
(wireless media)

Microwave Radio Wave Cellular Infrared Satellite



❖ Types of Transmission Media



1. Guided Media / Bounded Media

- In guided media, data signals flow through the help of wires / cables.
- Data is transmitted through these wires to a particular patch (पथ/रास्ता).
- These wires are made of copper, tin or silver.
- Guided media also known as Bounded media, which are those that provide a conduit (नलिका) from one device to another device.

➤ Bounded transmission media / Guided Media are three types

A. Twisted-pair cable

B. Coaxial cable

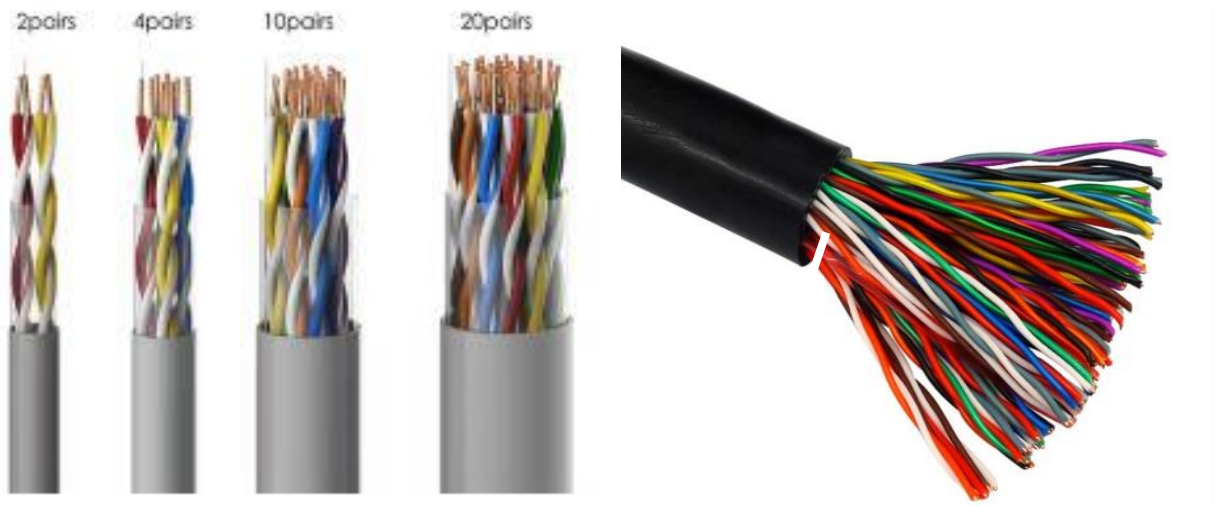
C. Fiber-optic cable

A. Twisted Pair Cable/wire

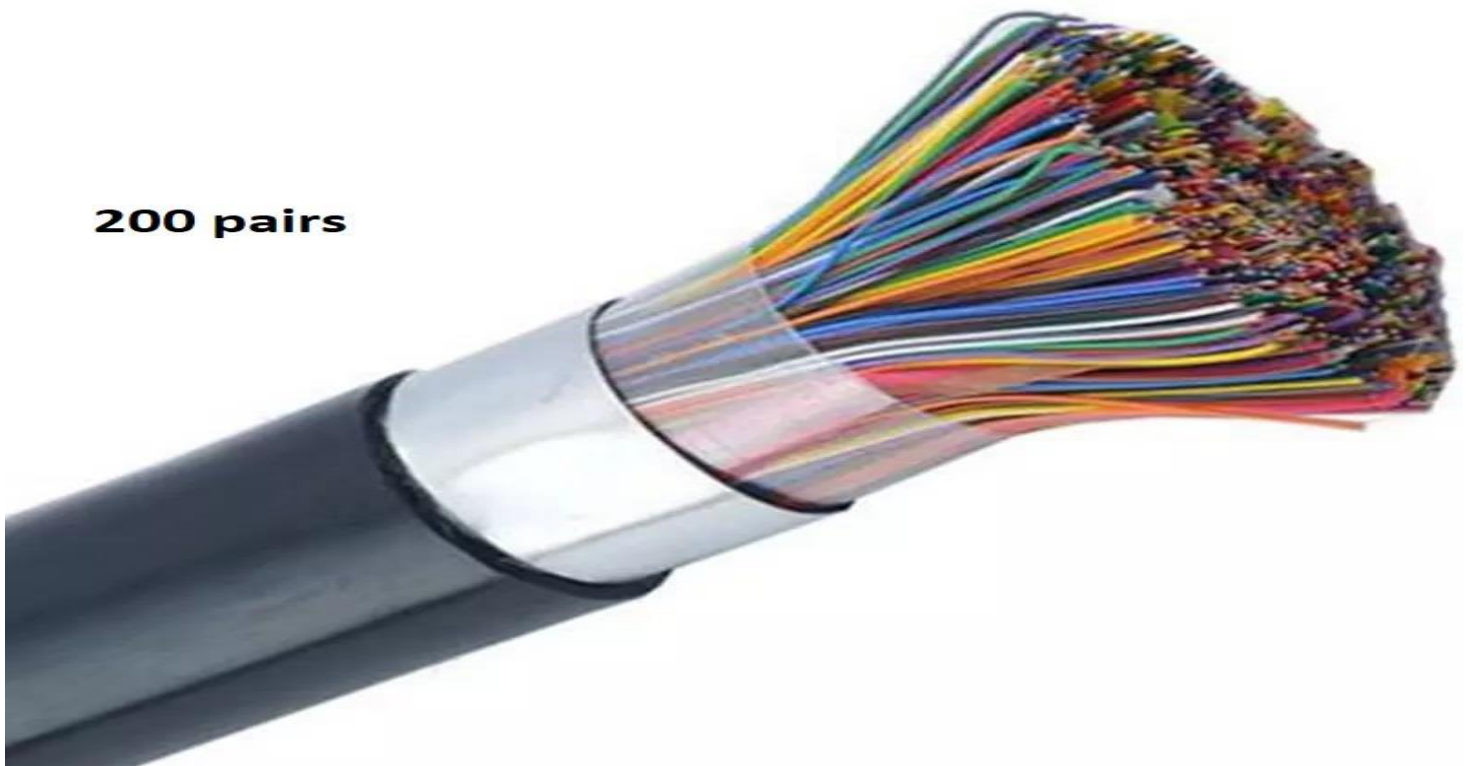
- The wires are twisted (मुड़ा) together in pairs (जोड़ा).
- The less expensive and most widely used guided transmission medium.
- Twisted pair (TP) maybe used to transmit both analog and digital signal.
- It is lightweight, cheap, can be installed easily, and they support many different types of network.
- Over longer distances, cables may contain hundreds of pairs.
- When data are transferred through this cable, the signal becomes weak after some distance. To maintain the signal, amplifiers are used for analog signals every 5 to 6 km, while repeaters are used for digital signals every 2 to 3 km.
- An amplifier increases the strength of a signal, whereas a repeater regenerates the signal before forwarding it.
- Example of Twisted pair cable : Ethernet cable (CAT5e/CAT6 / CAT6A) , telephone cable (CAT3), etc... (Here CAT means = Category)
- CAT5e, CAT6, Wire used for – Ethernet/LAN networking, Telephone line, IP camera, VoIP phone, PoE (Power over ethernet)

- DSL is not a type of cable; it is a technology in which the same telephone cable is used for the Internet.

TELEPHONE CABLE



200 pairs



DSL vs. Ethernet vs. Fiber Cable

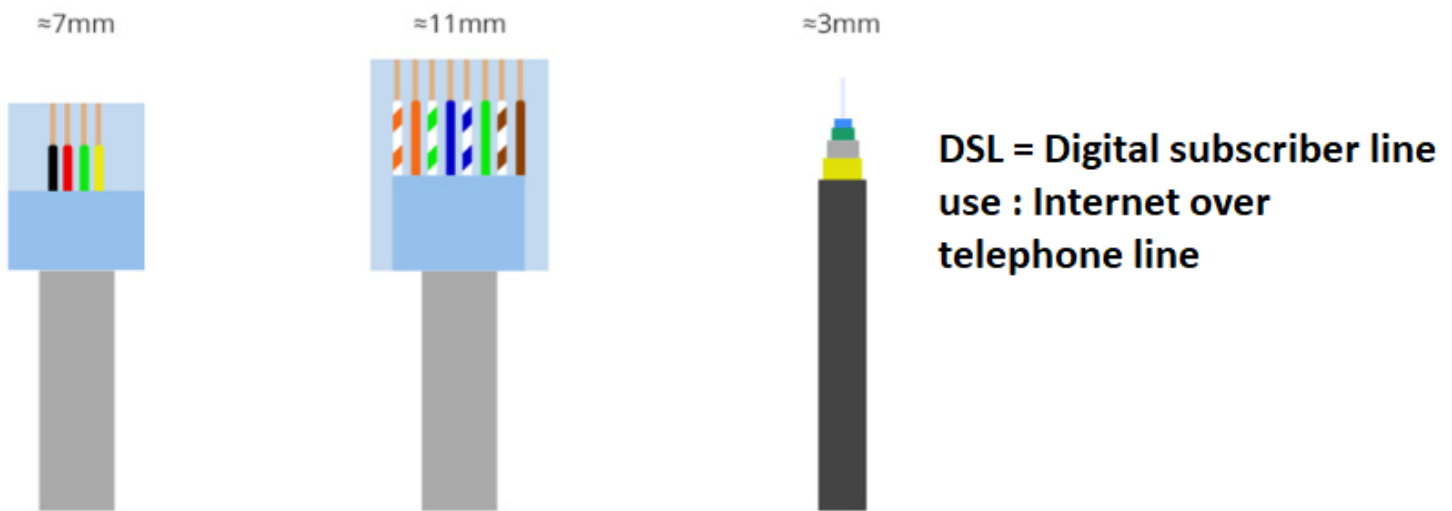


Figure: DSL vs Ethernet cable vs fiber cable

Practical

Image link 1 (repair cable)

[https://images.tribuneindia.com/cms/gall_content/2018/9/2018_9\\$largeimg15_Saturday_2018_005100203.jpg](https://images.tribuneindia.com/cms/gall_content/2018/9/2018_9$largeimg15_Saturday_2018_005100203.jpg)

Image link 2 (1000 pairs) : https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcRAL_g7tVqqBeviSq0GYOqntKkjqUCcRJkRz-kiOKJiUg&s

https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcRAL_g7tVqqBeviSq0GYOqntKkjqUCcRJkRz-kiOKJiUg&s

Image link 3 (old telephone distribution) :

https://live.staticflickr.com/5250/5251322429_c849ce1b55_b.jpg

Application/Use

- ✓ It is mainly used in short distance communication like LANs and phone lines.
- ✓ It is the most commonly used medium of telephone / land line.
- ✓ In the telephone system individual residential telephone sets are connected to the local telephone exchange or to end office by twisted pair wire.

- ✓ It is also commonly used within a building for LAN(Local area network) / Ethernet.

➤ Two types of Twisted pair cable

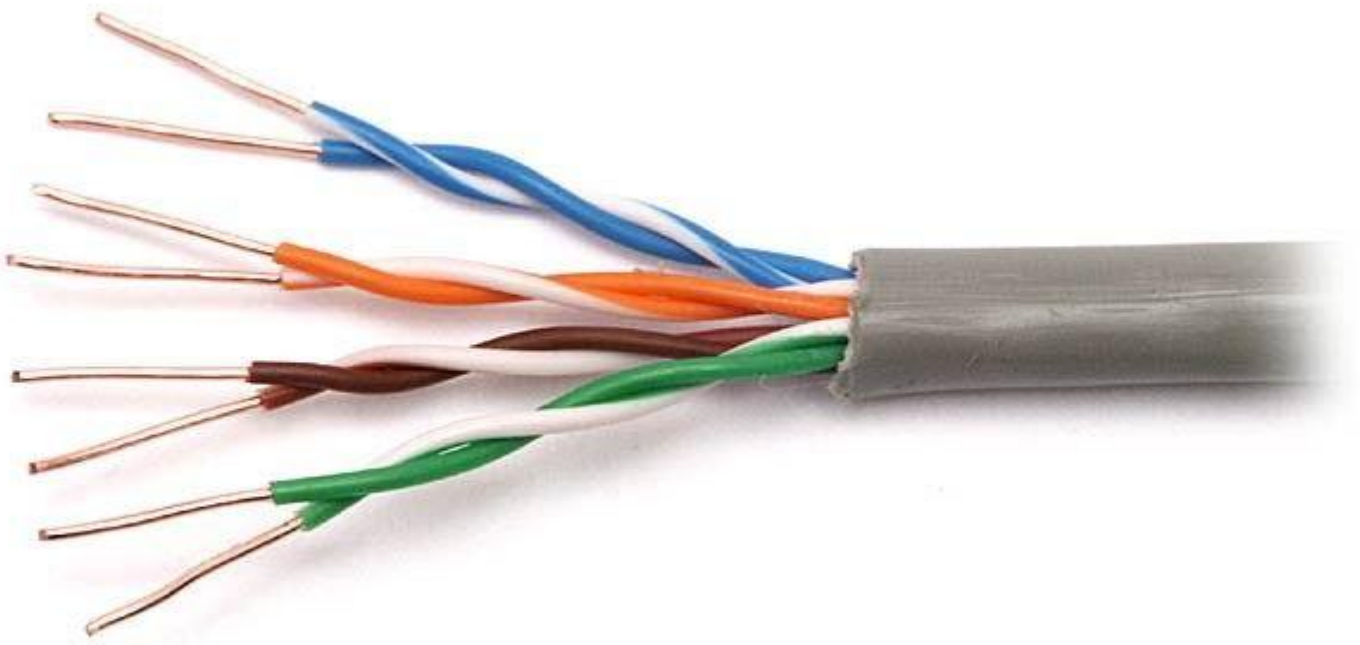
- i) **Unshielded twisted pair (UTP)**
- ii) **Shielded twisted pair (STP)**



i) **UTP (Unshielded twisted pair)**

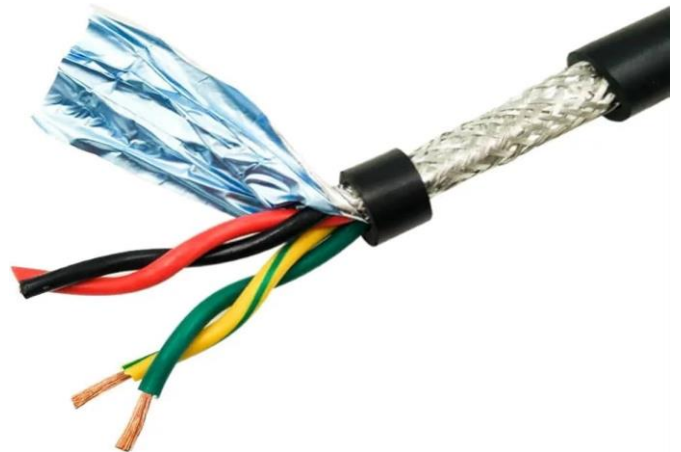
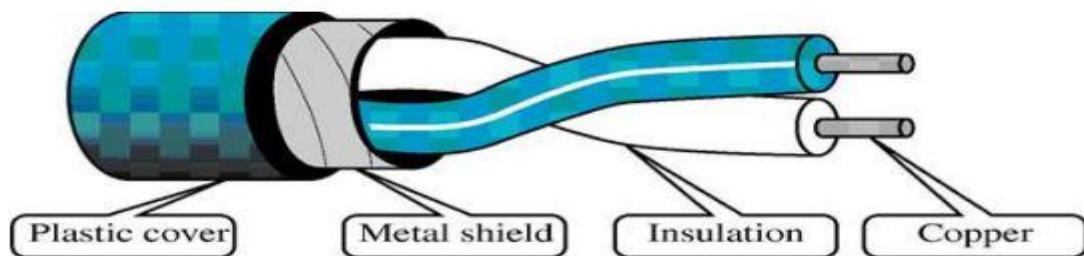
- Usually consists of two copper wires wrapped in individual plastic insulation.
- UTP cables are the most common telecommunications medium.
- The frequency range of the twisted pair cables enable both voice and data transmission.
- UTP cables consist of 2 or 4 pairs of twisted cable. Cable with 2 pair use RJ11 (Telephone) connector and 4 pair cable use RJ-45(PC/Laptop/router,...) connector.





ii) STP (Shielded twisted pair)

- The only difference between STP and UTP is that STP cables have a shielding in usually of aluminum or polyester material between the outer jacket and wire.
- The metal mesh around the insulated wires eliminates crosstalk. Crosstalk occurs when one line picks up some of the other signals traveling down another line.



➤ **Difference B/W UTP & STP**

Parameters	UTP	STP
Full Form	Unshielded Twisted Pair	Shielded Twisted Pair
Structure	cable with wires that are twisted together.	Twisted pair cable enclosed in foil / shield.
Cost	Cheaper than STP	Costlier than UTP
Weight	Lighter than STP	Heavier than UTP
Noise & interference	Prone to Noise and interference nature	Less prone to noise and interference
Data Speed	Supports slower speed than on STP	Support higher speed than UTP
Grounding of cable	Not required	Required
Target deployments	Locations less prone to interference like offices and homes.	Locations prone to interference like factories and airports

Extra Topic (Not in Syllabus)

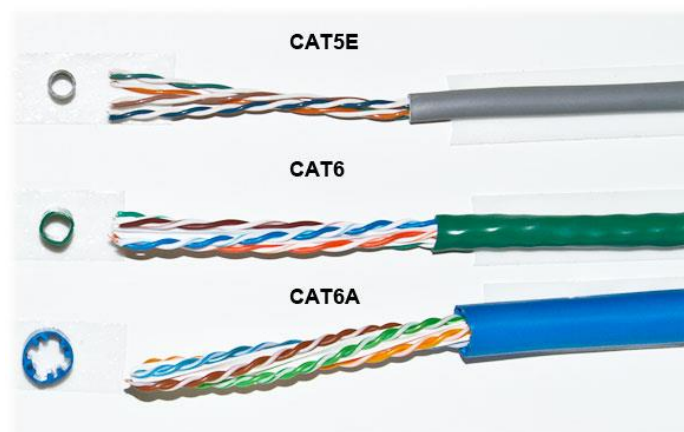
Twisted Pair cable			
Cable name	Connector name	use	Speed
CAT3	RJ 11	Telephone line	
CAT5	RJ45	Ethernet LAN, Internet	10-100mbps
CAT5E	RJ45	Ethernet LAN, Internet	1GBps
CAT6 / CAT6A	RJ45	Ethernet LAN, Internet	1GB – 10GBps
CAT6A	RJ45	Ethernet LAN, Internet	10GBps



RJ45



RJ11



RJ45 LAN Cable Wiring Color Code (T568A & T568B)

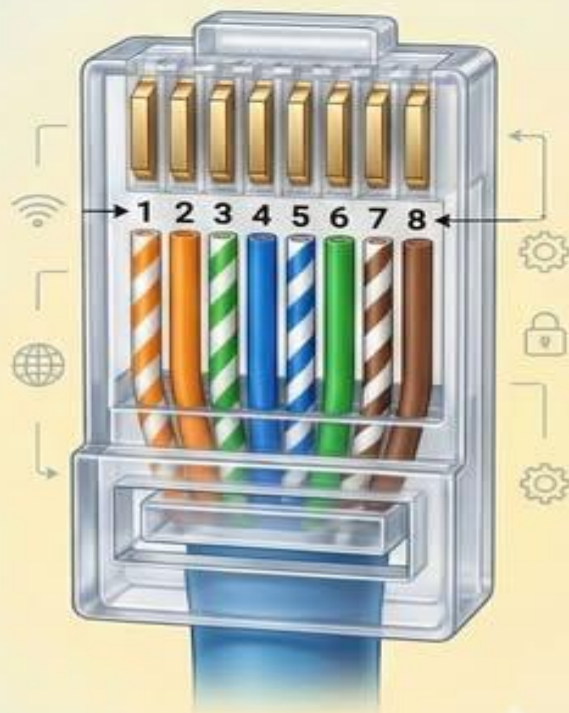
T568A Standard



T568A Color Code

Pin 1	White-Green
Pin 2	Green
Pin 3	White-Orange
Pin 4	Blue
Pin 5	White-Blue
Pin 6	Orange
Pin 7	White-Brown
Pin 8	Brown

T568B Standard



T568B Color Code

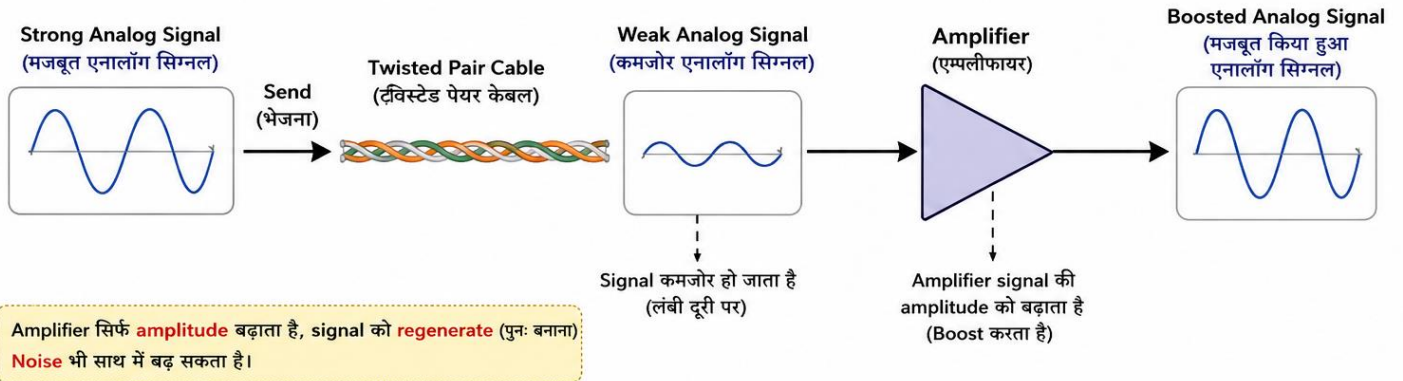
Pin 1	White-Orange
Pin 2	Orange
Pin 3	White-Green
Pin 4	Blue
Pin 5	White-Blue
Pin 6	Green
Pin 7	White-Brown
Pin 8	Brown

One Wrong Wire = No Internet



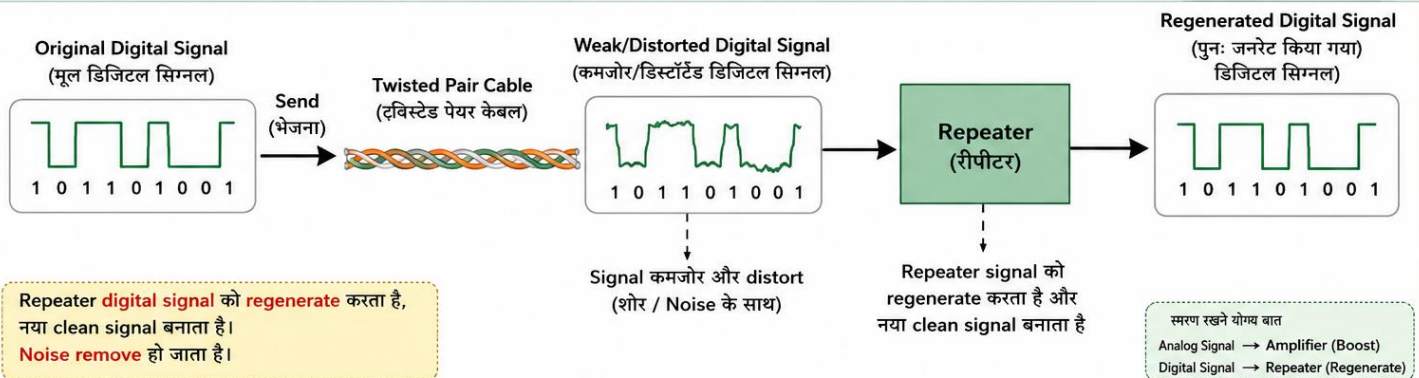
AMPLIFIER (एम्पलीफायर)

Amplifier **analog signal** को **boost** (मजबूत) करता है, उसकी **amplitude** बढ़ाता है लेकिन **signal** का **format** वही रहता है।



REPEATER (रीपीटर)

Repeater **digital signal** को **regenerate** (पुनः बनाता) करता है और नया, साफ और original signal बनाकर आगे भेजता है।



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